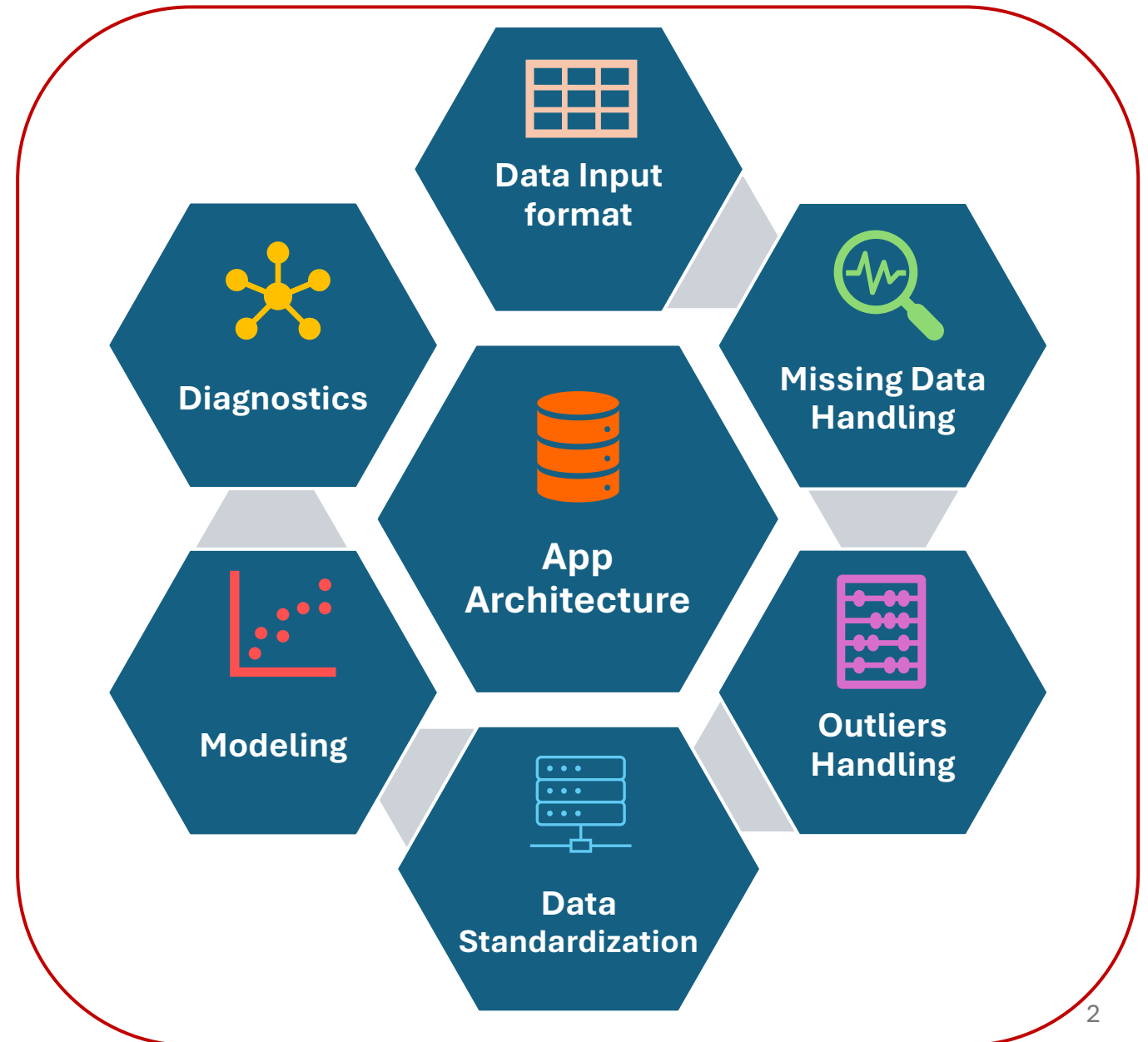
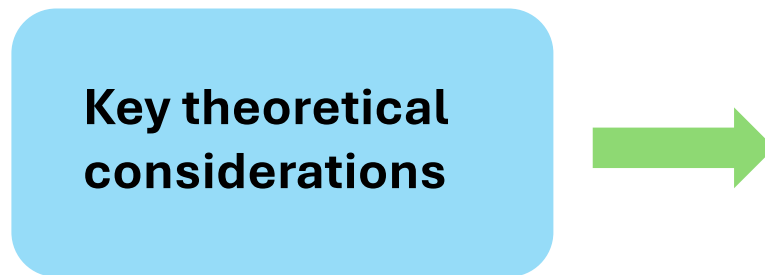

FlexibleGLMM

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Input data requirements



- Single file with complete raw data
- Format – excel or csv file
- Header Row with variable names without spacing (left_hypothalamus)
- Leave missing data as empty cells (blank – no value)
- Column variable order (small letters) :
 - subj, dv, iv, covar1, covar2, covar3, dv2, iv2, dv3, iv3.....

Wide Format

Team	Points	Assists	Rebounds
A	88	12	22
B	91	17	28
C	99	24	30
D	94	28	31

For GLM

Long Format

Team	Variable	Value
A	Points	88
A	Assists	12
A	Rebounds	22
B	Points	91
B	Assists	17
B	Rebounds	28
C	Points	99
C	Assists	24
C	Rebounds	30
D	Points	94
D	Assists	28
D	Rebounds	31

For GLMM

THE AMERICAN STATISTICIAN
2018, VOL. 72, NO. 1, 2–10
<https://doi.org/10.1080/00031305.2017.1375989>



OPEN ACCESS



Data Organization in Spreadsheets

Karl W. Broman^a and Kara H. Woo^b



Method	Advantages	Disadvantages
Listwise Deletion (Complete Case Analysis)	Reproducible results	Discards potentially useful data Reduced statistical power
Pairwise Deletion (Available Case Analysis)	Utilizes all available data	Inconsistent sample sizes Difficult to reproduce results

- **Listwise Deletion** - Large dataset with low missingness (<5%) under MCAR with high correlations
- **Pairwise Deletion** - Exploratory analyses under MCAR with low correlations



Removing outlier data points could **improve the robustness of inferences**

Method	Type	Use Case	Robustness
Standard Deviation (Z-score)	Univariate	Extreme values (normal data)	Low (Sensitive to outliers & non-normality)
Cook's Distance	Regression Influence	Influential observations	Low (Influenced by extreme points)
Mahalanobis Distance	Multivariate	Covariance-based outliers	Moderate (Improves with robust covariance)
Lookout (LOOCV KDE)	Density-based	High-dimensional anomaly detection	High (Distribution-free)
DHARMa Residuals	Simulation-based	GLMM diagnostics	High (Robust in complex, high-dim data)

- Look for errors in the data (Use scatterplot)
- Check data range (min-max)
- Fit a preliminary model and plot your residuals

Biological and Experimental context is important in deciding outliers

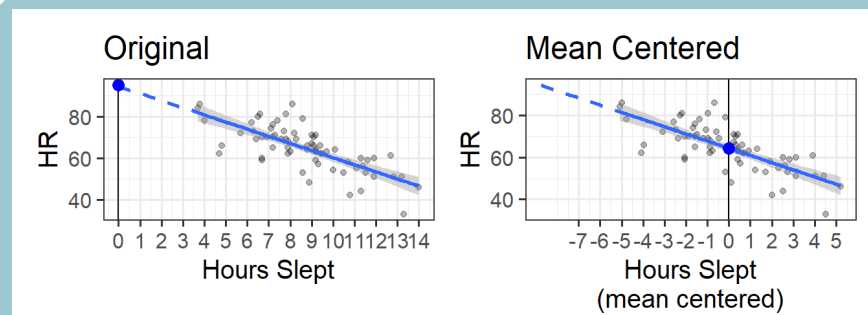


These preprocessing steps improve **interpretability**, **numerical stability**, and **model performance**.

Mean Centering

Subtracting the mean from each variable

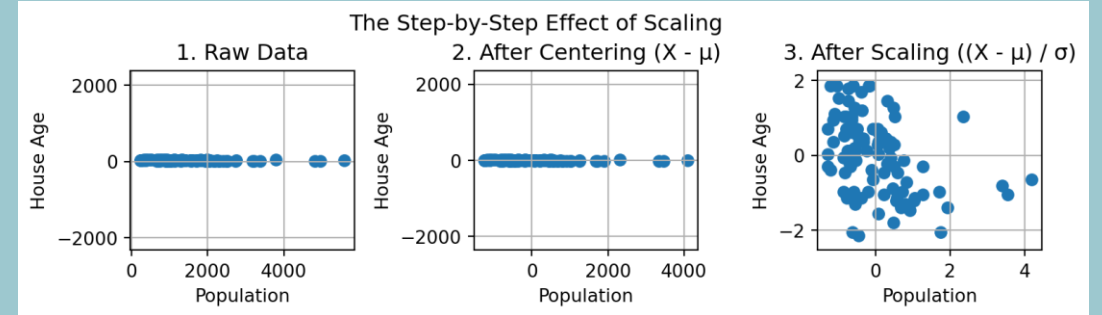
- Reduces multicollinearity
- Improves numerical stability
- Interpretability of the intercept



Scaling (Z-scoring)

Dividing by standard deviation

- Makes coefficients comparable
- Improves optimization and convergence
- Essential for regularized regression



Understanding Your Data



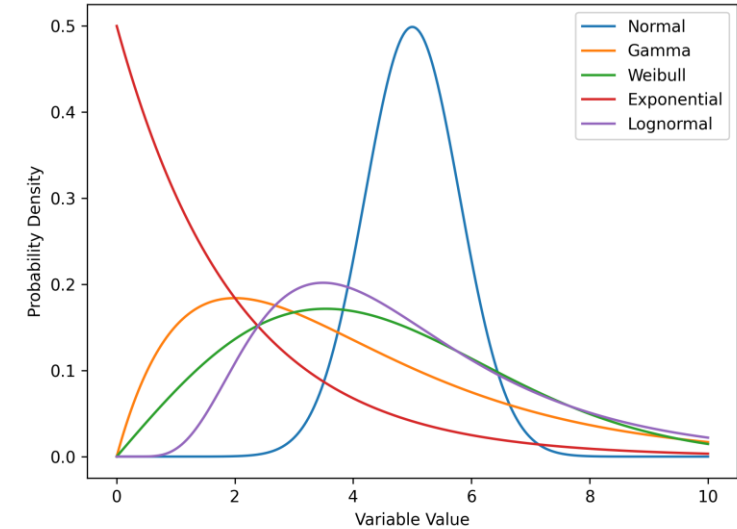
Data = Variable **Structure** (Type) + Probability **Distribution**

Structure

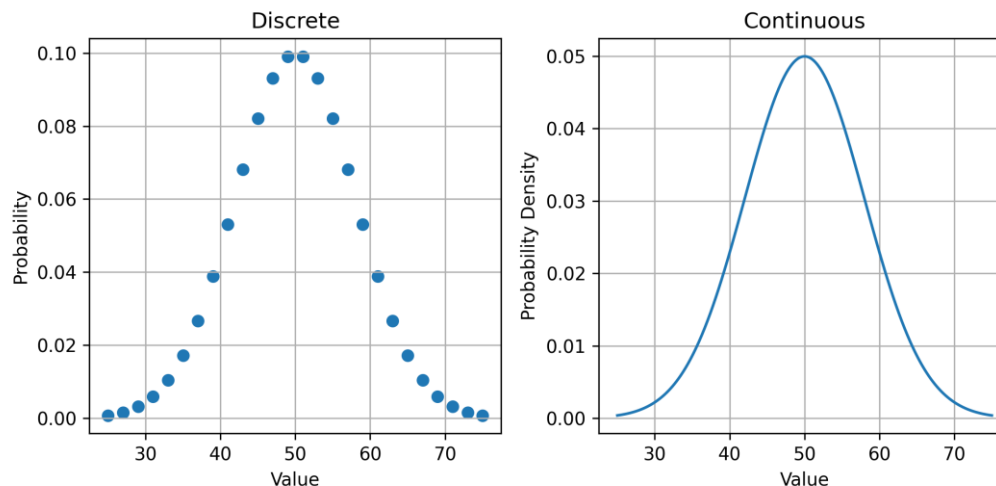
- Continuous variables
- Count variables
- Categorical variables
- Binary variables

Distribution

- Normal
- Gamma
- Lognormal
- Binomial



Variable Structure with Distribution



Structure tells us what data *can be*;
Distribution tells us what data *actually is*.

Most model failures happen when:

- **Structure stays the same**
- **Distribution shifts**

Understanding Your Data



Before modeling, deeply understand what you're working with.

1

Outcome

What are you predicting?

2

Exposure/Predictor

What are your key predictors?

3

Covariates

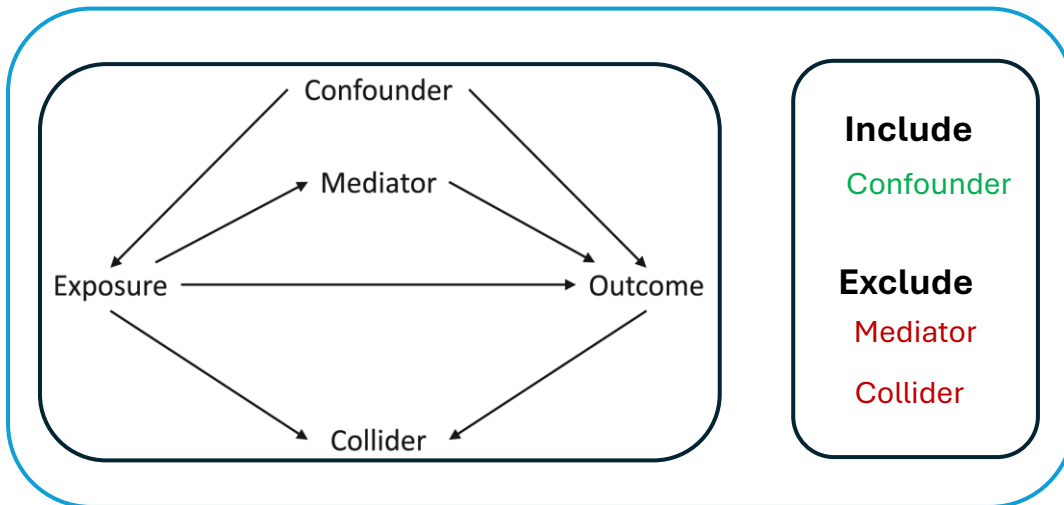
What are your covariates?

4

Assumptions

What must hold for valid inference?

A covariate can be a confounder, a collider, or a mediator.



Statistical tests are based on certain assumptions

Reliable model, accurate predictions, unbiased coefficient estimates, and valid statistical inferences

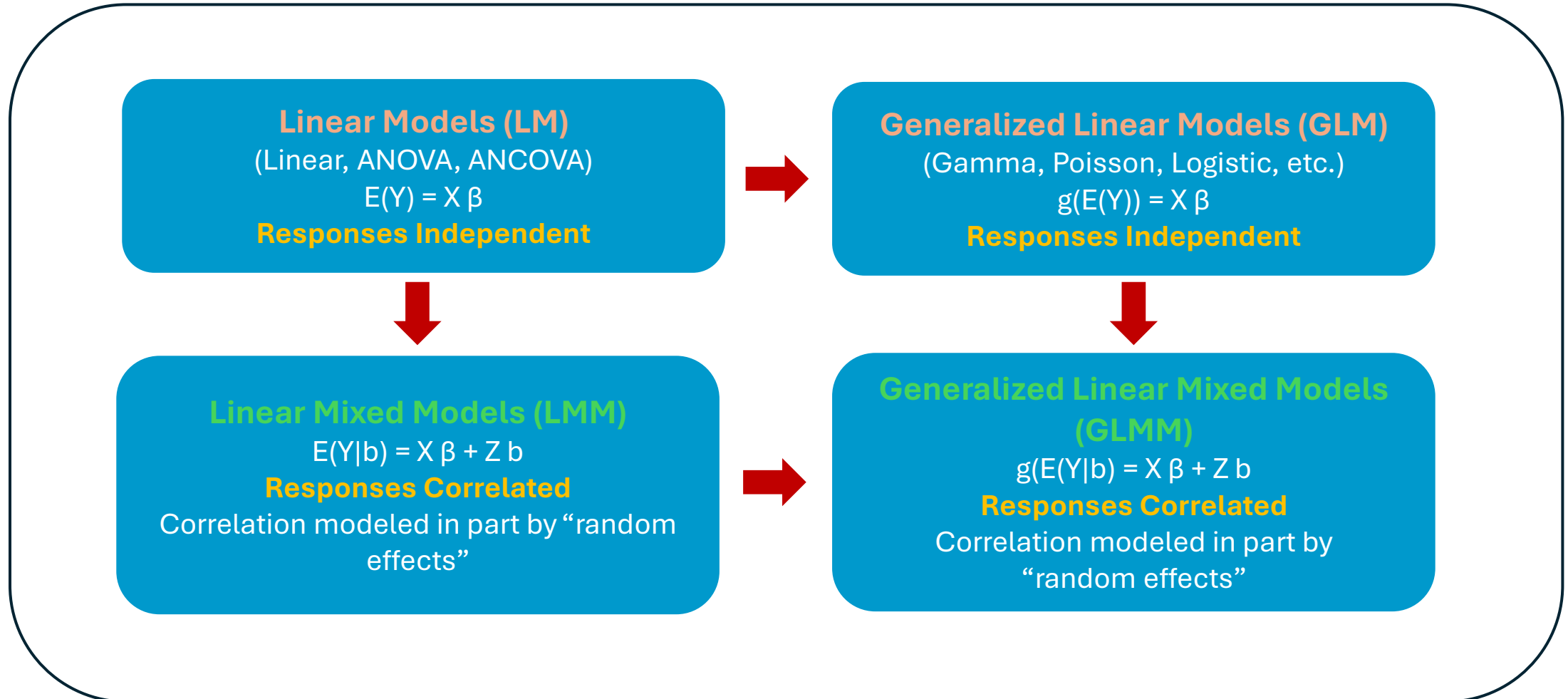
***Assumptions are not footnotes.
They are the foundation.***

Why Assumptions are important?



Scenario	Standard Test (Assumptions Hold)	Key Assumptions	Alternative Test (Assumptions Relaxed)	Why Alternative?
Compare means of 2 independent groups	Student's t-test	Normality, equal variances , independence	Welch-Satterthwaite t-test	Handles unequal variances
Compare means of >2 groups	One-way ANOVA	Normality, equal variances	Welch ANOVA	Robust to unequal variances
Post-hoc after ANOVA	Tukey HSD	Equal variances	Games-Howell test	Unequal variances
Equality of variances	F-test	Normality	Levene / Brown-Forsythe	Robust to non-normality

Statistical tests are tools, not ornaments.



Mixed Models (SAS vs R)



Feature	SAS PROC GLIMMIX	R lme4	R nlme	R afex
Primary scope	Linear + generalized mixed models	Linear + generalized	Primarily LMM	ANOVA wrapper (lme4)
Estimation method	REML/ML for LMMs; PQL for GLMMs	REML/ML for LMMs; Laplace /AGQ for GLMM	REML/ML	Uses lme4
Handles complex covariance structures?	Extensive	Only random effects	For LMM only	Only random effects

SAS code:

- Penalized quasi-likelihood (PQL) **[Default]** : `proc glimmix data=test method=RSPL`
- LAPLACE : `proc glimmix data=test METHOD=LAPLACE`

Shiny App integrates all 3 packages providing all features and model comparison options

Random effect structure in GLMM



Converting random effect expression from SAS to R

SAS:

random intercept color size

/type = vc subject = subject solution;

R:

formula ((1|subject) + (0 + color|subject) + (0 + size|subject))

Subject – categorical grouping variable

Both color & size - continuous variables

*type = vc gives **uncorrelated** random slopes*

Shiny App provides a helper table

Formula (lme4)	Alternative (lme4)	Formula (nlme)	Meaning
(1 g)	-	~ 1 g	Random intercept with fixed mean
(1 g1:g2)	-	~ 1 g1:g2	Random intercept for each level of the interaction between g1 and g2
(1 g1/g2)	(1 g1) + (1 g1:g2)	~ 1 g1/g2	Intercept varying among g1 and g2 within g1 (g2 Nested within g1)
(1 g1) + (1 g2)	-	-	Intercept varying among g1 and g2 (Crossed random intercepts)
(x g)	(1 + x g)	~ x g	Correlated random intercept and slope
(x g)	(1 g) + (0 + x g)	list(g = pdDiag(~ x))	Uncorrelated random intercept and slope
(0 + x g)	-	~ x - 1 g	Uncorrelated random slope/Random slope for x only (no intercept)

g, g1, g2 are **categorical grouping variables**, while **x** is typically a **continuous predictor**.

Covariance structures in GLMM



SAS	R nlme	Description
AR(1)	corAR1(correlated errors)	Balanced repeats
CS	corCompSymm	Ordered trials
UN()	corSymm	unbalanced data

Requirements for fitting covariance structures in GLMM

- grouping variable should be present in random effect
- grouping variables should be **factors/categorical** variables (as.factor)
- ≥ 2 observations **per group** to estimate within-group correlation
- ≥ 2 **groups** for random-effect variance estimation
- correlation grouping level $\leq$ random grouping level (for AR1 etc.)
- correlation structure should match **data ordering** (for AR1 etc.)
- requires explicit **within-group index/ordering** variable (for AR1 etc.)
- time(ordering) variable (if present) should be **numeric** (for AR1 etc.)

Throws errors or warnings

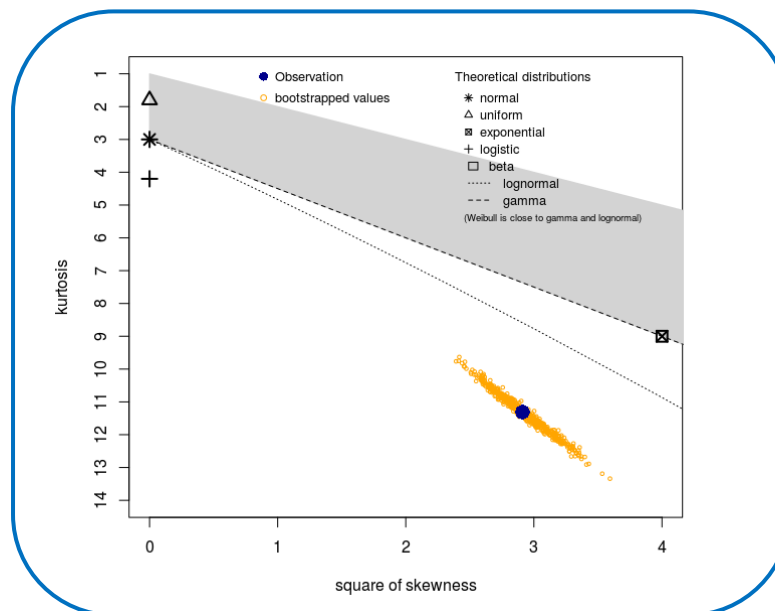
Confusing?

Shiny App provides option to auto-select covariance structure

Fit Distributions and Goodness-of-fit



Cullen and Frey or skewness-kurtosis plot

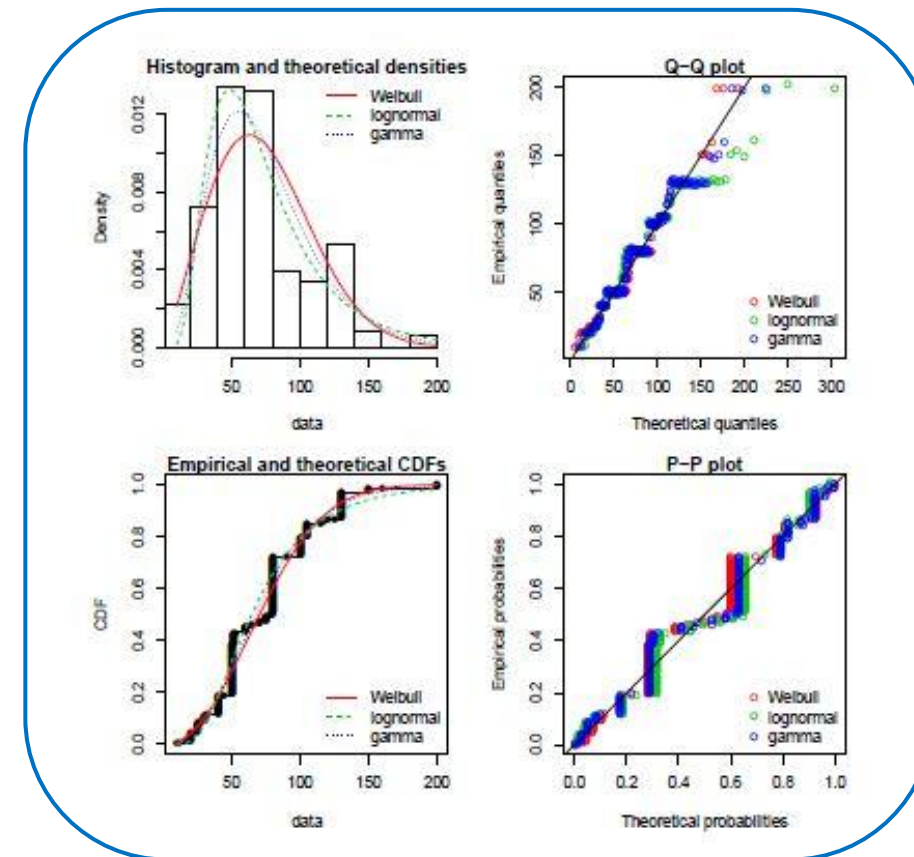


Goodness-of-fit statistics

Statistic	Focus	Key Strength
Kolmogorov-Smirnov (KS)	Maximum difference	Widely used, robust
Cramer-von Mises (CvM)	Integrated squared differences	Better overall fit
Anderson-Darling (AD)	Weighted tail differences	Highest power for tails

Fit distribution tells about **unconditional distribution shape** whereas GLMM care about **conditional residual distribution**.

Compare the distributions



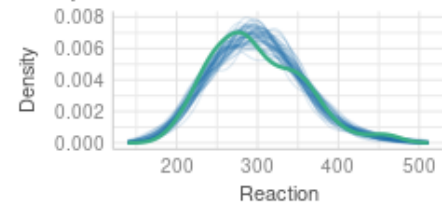
Check residual diagnostic plots

Model Diagnostics



Posterior Predictive Check

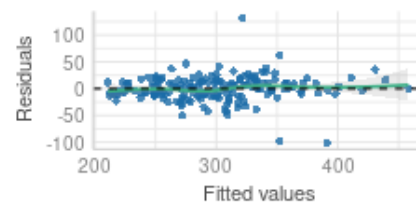
Model-predicted lines should resemble observed data



— Model-predicted data — Observed data

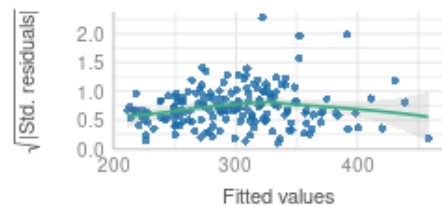
Linearity

Reference line should be flat and horizontal



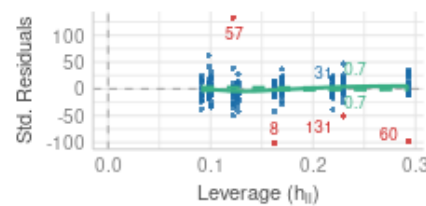
Homogeneity of Variance

Reference line should be flat and horizontal



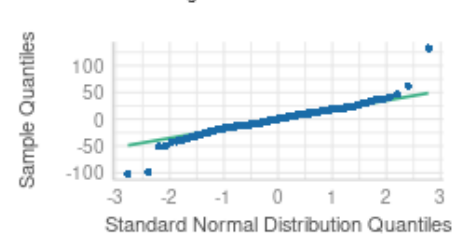
Influential Observations

Points should be inside the contour lines



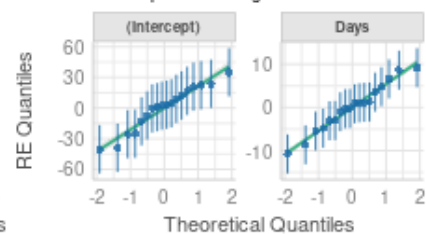
Normality of Residuals

Dots should fall along the line



Normality of Random Effects (Subject)

Dots should be plotted along the line



Indices of model performance

AIC	AICc	BIC	R2 (cond.)	R2 (marg.)	ICC	RMSE	Sigma
159.0	159.3	171.1	0.933	0.303	0.904	0.373	0.378

Singular Fit issue

Tells about Model
overfitting

Convergence failure

Scale and center
predictors

**Shiny App provides plots, indices along with singular fit
& convergence failure check**



Understanding the Nuances Beyond the Overall Effect

Why It Matters

- ❑ **Pinpoint Specific Differences**
Identifies which pairwise group comparisons are significant.
- ❑ **Account for Model Complexity**
Adapts to non-normal families (binomial, Poisson) and random effects like repeated measures or nested structures.
- ❑ **Avoid Misinterpretation**
Only a subset of comparisons may drive an overall significant effect.

Post-Hoc Tests

Tukey's honestly
significance difference for pairwise
comparisons for balanced designs

Sidak-Bonferroni
for a small number of groups

*Shiny App automatically switches to Sidak if
number of groups are small.*

GLMM Shiny App (Flexible GLMM) Demo



sleepstudy — modified from the R package lme4

Data Structure (N=400):

- Repeated measures (10 days per subject)
- Continuous outcome (Reaction_time)
- Continuous time predictor (Days)
- Subject-level grouping (Subject)
- Covariates (Age, BMI, TST, Chronotype & Caffeine)

<https://puneet-talwar.shinyapps.io/FlexibleGLMM/>

